

PLC Architecture and Parts

1 Typical PLC Parts

Core Components

A typical PLC system consists of four primary sections that work together to execute control logic:

- **Central Processing Unit (CPU):** The brain of the PLC. It executes the control program, performs logic calculations, and manages data flow between other components.
- **Input/Output (I/O) Section:** The interface between the PLC and the outside world.
 - **Inputs:** Receive signals from sensors, switches, and transmitters.
 - **Outputs:** Send control signals to actuators, motors, and lights.
- **Power Supply:** Converts the available incoming AC power to the DC power required by the CPU and I/O modules.
- **Programming Device:** The tool used to develop, download, and monitor the control program (e.g., a laptop or a dedicated handheld programmer).

2 PLC Architecture

Defining Architecture

In the context of PLCs, **Architecture** can refer to the hardware design, the software structure, or the combination of both.

2.1 Open vs. Closed Architecture

Feature	Open Architecture	Closed Architecture
Interconnectivity	Allows the system to be connected easily to devices from various vendors.	Design is proprietary; difficult to connect to systems from other manufacturers.
Compatibility	Supports programs and modules made by different manufacturers.	Limited to products and protocols from a specific vendor.
Components	Uses widely available "off-the-shelf" components.	Uses specialized, vendor-specific parts.

Trend Towards Openness

Modern industrial systems increasingly favor **Open Architecture** to reduce costs, increase flexibility, and ensure long-term support through standardized protocols.

Understanding the architectural foundation is key to scaling industrial control systems.